

A Factor-Structured, Regime-Conditional GAN for Multi-Asset Synthetic Financial Data

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Abstract

Generating realistic synthetic multi-asset financial returns is essential for stress testing, scenario analysis, and risk management—applications that require a generator to behave sensibly across distinct macro regimes and to preserve the within-sector dependence structures that drive real portfolio risk. We propose **SF-MarketGAN-R**, whose contributions are three-fold. (i) *Sector-augmented factor decomposition*: we combine 5 global PCA factors with 5 custom Fama–French-style stock industry factors, 4 S&P GSCI commodity sub-index factors, and 2 bonds level/slope factors, then apply sequential Gram–Schmidt orthogonalisation and unit-variance standardisation, yielding 16 mutually orthogonal factors that specifically capture same-sector co-movement (e.g. Visa–Mastercard in FINANCE, Gold–Silver in PRECIOUS) missed by prior PCA-only designs. (ii) *Factor-structured capacity restriction*: the generator fixes factor loadings to rolling OLS and residuals to a training-sample correlation Cholesky, restricting adversarial learning to macro-conditioned corrections on intercepts and volatilities. (iii) *Regime-gated correction-magnitude schedule*: the (α, σ) -correction envelope is gated by a VIX-aligned encoder so the model trusts OLS in calm regimes and expands under stress. On a 60-asset universe (bonds, commodities, equities) with strictly lagged factor conditioning, SF-MarketGAN-R achieves out-of-sample correlation Frobenius 2.58 against factor-bootstrap 3.36 (−23.3%) and a training-residual-Cholesky bootstrap baseline FB+shrunk 3.12 (−17.4%), winning 5 of 6 evaluation metrics. The improvement is structural: autocorrelation, volatility-clustering, and leverage-effect stylized-fact scores all drop by 10–13% relative to both factor-bootstrap baselines, while same-sector pair rolling-correlation RMSE matches the best classical method on the Visa–Mastercard payments pair (0.091 vs FB+shrunk 0.084) without degrading aggregate metrics. We compare against 7 classical and neural alternatives (factor bootstrap with and without shrinkage, DCC-GARCH, Student- t copula, block bootstrap, MarketGAN-60, and our prior PCA-only variant) on six same-sector pairs and show that factor-bootstrap with an appropriately estimated residual Cholesky already matches a GAN’s pair-correlation fidelity, so the GAN’s incremental value is not on any single pair but on the *structural* stylized-fact fit that no classical method reaches.

1 Introduction

Synthetic financial data generation serves critical functions in quantitative finance: stress testing portfolios against scenarios that have never occurred, estimating risk metrics with larger sample sizes than history provides, and preserving privacy when sharing proprietary trading data. The challenge is acute for multi-asset portfolios, where generating realistic cross-asset dependence structures—correlations, tail co-movement, and volatility spillovers—is substantially harder than matching marginal distributions of individual assets.

Traditional approaches fall into two camps. Parametric models such as DCC-GARCH [Engle, 2002] impose strong distributional assumptions that often fail to capture the nonlinear, regime-dependent dynamics of real markets. Non-parametric methods like factor bootstrap [Huh et al.,

2026] preserve the empirical structure of historical data but cannot generate counterfactual scenarios conditioned on hypothetical macro states. Recent work on financial GANs, including MarketGAN [Huh et al., 2026] and related architectures [Vuletić et al., 2023, Cont et al., 2025], attempts to bridge this gap by learning the full return-generating process adversarially. These models typically learn corrections to factor loadings (β), idiosyncratic volatilities (σ), and residual dynamics (ε) simultaneously.

We find, however, that this full-learning approach systematically hurts OOS generalization. Ablations show two consistent patterns: (1) learned residual networks inject spurious cross-class correlation that does not persist out of sample, and (2) adversarial corrections to factor loadings distort the cross-asset covariance structure that rolling OLS already captures well. A third pattern emerges from a focused failure-mode study: purely data-driven (PCA) factor decompositions leave *same-sector* pair co-movement in the residual space, and any fixed-Cholesky residual prior then becomes a bottleneck for those pairs specifically. For instance, with only global + within-class PCA factors, the Visa–Mastercard training-period residual correlation is +0.57 (out-of-sample +0.73); a shrinkage-toward-identity residual prior pushes this to +0.17, catastrophically underestimating the payments co-movement that a risk analyst cares most about.

These observations motivate two coupled design choices: (i) a *sector-augmented, Gram–Schmidt-orthogonalised* factor set that puts the same-sector common component into $\hat{\beta}$ rather than the residual; and (ii) a *factor-structured generator* that restricts the adversarial component to quantities where classical estimators are demonstrably weak.

Our contributions.

- **Sector-augmented factor decomposition (representation contribution).** We extend the hierarchical PCA factor model of our prior MarketGAN work with 5 custom Fama–French-style stock industry factors (splitting FINANCE out of FF5’s “Other” so Visa/Mastercard/JPM/BRK.B share a bucket) and 4 S&P GSCI commodity sub-indices (ENERGY, PRECIOUS, INDUSTRIAL, AGRICULTURE), then apply sequential Gram–Schmidt orthogonalisation and unit-variance standardisation. The resulting $K = 16$ factor panel has training off-diagonal correlation 0 and β -magnitudes a factor of 40 smaller than the raw (non-standardised) equal-weight sector means, making rolling OLS numerically stable at $N/K \approx 3.8$. Ablations (§4.5) show this alone reduces OOS correlation Frobenius by 11% relative to the 11-factor PCA-only baseline.
- **Factor-structured generator (architectural contribution).** We restrict generator capacity to macro-conditioned corrections on intercepts (α) and idiosyncratic volatilities (σ) via a TCN backbone with FiLM conditioning. Factor loadings (β) are taken directly from rolling OLS. Residuals use a *fixed* training-sample correlation Cholesky L_ρ (not a shrinkage prior, not a learned network); Appendix A reports three failed attempts to make L_ρ adversarially trainable.
- **Regime-gated correction schedule (mechanism contribution).** A VIX-aligned encoder gates the magnitudes of the (α, σ) -correction heads: a tight bound ($\eta_\alpha = 0.01$, $\eta_\sigma = 0.05$) in calm regimes, relaxed to ($\eta_\alpha = 0.10$, $\eta_\sigma = 0.20$) under stress. The encoder is trained with an auxiliary BCE loss aligned to a hard VIX-z threshold, which we verify with a hard-threshold ablation.
- **Main 60-asset OOS.** Under strictly lagged factor conditioning (f_{t-1}), SF-MarketGAN-R achieves OOS correlation Frobenius 2.53 vs. factor-bootstrap 3.36 (−24.6%) and shrinkage-prior baseline 3.12 (−18.8%), winning 4 of 6 evaluation metrics. The three stylized-fact met-

rics (ACF, vol-clustering, leverage) each improve by 10–13% over the same-factor FB+shrunk baseline—a structural advantage that a better residual Cholesky alone cannot produce.

- **Same-sector pair-correlation baseline comparison.** We benchmark against 7 classical and neural alternatives on six same-sector pairs covering payments, precious metals, Treasuries, energy, mega-tech, and cross-subgroup. Factor-bootstrap with an appropriately estimated residual Cholesky matches SF-MarketGAN-R on all six pairs within measurement noise—so the GAN’s incremental value is *not* on pair fidelity but on structural stylized-fact fit that no classical method approaches.

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the model architecture and its theoretical motivation. Section 4 reports experimental results including the pair-correlation baseline study, stylized-fact comparison, and ablations. Section 6 discusses limitations and future directions.

2 Related Work

Multi-asset financial data generation. Generating realistic synthetic returns for multiple correlated assets has been approached from several angles. Classical methods include multivariate GARCH models [Engle, 2002, Bollerslev et al., 1988], copula-based generators [Marti, 2020, Marti et al., 2021], and factor bootstrap methods that combine rolling parameter estimates with resampled residuals. Among these, the factor bootstrap is surprisingly competitive: by decomposing returns into systematic (factor-driven) and idiosyncratic components, it preserves the cross-asset correlation structure without learning. Our work uses the factor bootstrap as the primary baseline and shows that a GAN can improve upon it—but only when the GAN’s learning is carefully constrained.

GANs for financial time series. Several recent works apply GANs to financial data. MarketGAN [Huh et al., 2026] introduces a factor-structured generator with TCN backbone and WGAN-GP training, learning corrections to rolling OLS estimates of α , β , and σ simultaneously. Fin-GAN [Vuletić et al., 2023] focuses on supervised probabilistic forecasting for individual series. Tail-GAN [Cont et al., 2025] targets tail-risk functionals via VaR/ES-shaped critics. CorrGAN [Marti, 2020] and cCorrGAN [Marti et al., 2021] generate correlation matrices directly in the elliptope. Our approach differs from all of these by demonstrating that the *residual learning* component of financial GANs—typically considered essential—actually hurts OOS correlation fidelity. We achieve better results by removing it entirely and using fixed residual covariance instead.

Factor models in asset pricing. Our architecture builds on the factor model tradition in finance. The APT [Ross, 1976] posits that returns are driven by a small number of common factors plus idiosyncratic noise. Conditional factor models [Kelly et al., 2019a] allow time-varying loadings and risk premia. Stochastic volatility factor models [Pitt and Shephard, 1999, Lopes and Carvalho, 2007] extend this to time-varying factor and idiosyncratic variances. Our generator can be viewed as a parsimonious stochastic volatility factor model where the volatility dynamics are learned adversarially from macro covariates rather than specified parametrically.

GAN training stability. Training stability is a well-known challenge for GANs. WGAN-GP [Gulrajani et al., 2017] uses gradient penalty for Lipschitz continuity, while spectral normalization [Miyato et al., 2018] constrains the discriminator’s spectral norm. BigGAN [Brock et al., 2019] uses orthogonal regularization to prevent mode collapse. R1 gradient penalty [Mescheder et al.,

2018] offers an alternative to WGAN-GP. We adopt SN-GAN with hinge loss, which constrains the discriminator through spectral normalization alone and trains reliably at $n_{\text{critic}} = 1$ in our setting.

Crisis transmission and correlation dynamics. The literature on financial contagion informs our architectural choices. Forbes and Rigobon [2002] show that apparent crisis contagion largely reflects continuation of existing linkages through common channels, supporting our decision to route macro stress through factor corrections rather than residual networks. Longin and Solnik [2001] demonstrate that correlations increase in bear markets specifically, not just in high-volatility periods. These findings motivate our macro-conditioned volatility correction, which adjusts idiosyncratic risk based on current macro state.

3 Method

3.1 Factor Model Decomposition

We decompose daily log returns $r_t \in \mathbb{R}^N$ for $N = 60$ assets as:

$$r_{i,t} = \hat{\alpha}_{i,t} + \hat{\beta}_{i,t}^\top f_t + \hat{\sigma}_{i,t} \cdot \varepsilon_{i,t} \quad (1)$$

where $f_t \in \mathbb{R}^K$ are observable factors, $\hat{\alpha}_{i,t}$, $\hat{\beta}_{i,t}$, and $\hat{\sigma}_{i,t}$ are rolling OLS estimates (126-day past window), and $\varepsilon_{i,t}$ is the standardised residual.

Sector-augmented factor set ($K = 16$). We construct $K = 16$ factors in four tiers, combining data-driven global structure with hand-labelled industry factors grounded in standard asset-pricing taxonomies:

1. **5 global PCA factors.** PCA on the full N -asset return matrix fitted on training data only. Captures shared dynamics across all three asset classes (explained variance 58.4% on training).
2. **2 bonds class-PCA factors.** PCA on bonds rows of the global-factor residuals. Retained only for bonds because fixed-income returns are driven by the two-factor level/slope structure of Litterman and Scheinkman [1991], not by a GICS-style industry taxonomy (explained variance 76.6% on training).
3. **5 stock industry factors** (custom 5-bucket, Fama–French-inspired). We use HiTEC, HEALTH, FINANCE, CONSUMER, and ENERGY, assigning each of our 20 equities to exactly one bucket. The bucket boundaries follow Fama–French FF5 with one substantive deviation: we split FINANCE out of FF5’s “Other” category so that payments processors (Visa, Mastercard) share a loading with banks (JPM) and diversified financials (BRK.B). Each factor is the equal-weight mean return of its bucket.
4. **4 commodity GSCI sub-index factors.** Following S&P GSCI methodology we group the pure single-commodity names into ENERGY, PRECIOUS, INDUSTRIAL, and AGRICULTURE; diversified commodity ETFs (DBC, GSG, DJP, PDBC, FTGC) are intentionally left unassigned and pick up multi-factor exposure through the OLS fit.

Sequential Gram–Schmidt orthogonalisation and standardisation. Raw equal-weight sector returns are strongly collinear both within-tier and with the global PCA factors. We process the 16 raw columns in priority order [global, bonds, stock sector, commodity sector]; for each column $k \geq 1$ we fit an intercept-inclusive OLS of column k on columns $0, \dots, k-1$ using *training-period data only*, subtract the fitted values from the full series, and continue. A final step divides each column by its training-period standard deviation. On training data the off-diagonal correlation among the 16 factors is 0 by construction; on the held-out test window all pairwise $|\text{corr}|$ remain below 0.05. Standardisation produces β -magnitudes ($|\hat{\beta}| \approx 0.002$ in mean, ≤ 0.033 in max) that are numerically stable for rolling OLS on a 126-day window with $N/K \approx 3.8$.

Why sector factors help OOS pair-correlation. The industry factors specifically target same-sector pair correlations that the previous 11-factor hierarchical PCA design missed. Under our $K = 11$ PCA specification, the Visa–Mastercard residual correlation on training was +0.569 (test: +0.726) because stocks class-PCA captures only generic market/size directions, not payments-industry co-movement. Under the 16-factor sector specification, the same pair’s training residual correlation drops to +0.249 (−56%) and test residual correlation to +0.448 (−38%); the payments signal is now loaded onto FINANCE’s $\hat{\beta}$ rather than sitting in the residual space, where a fixed Cholesky prior cannot track it. Analogous reductions hold for Gold–Silver (+0.424 $\rightarrow$ +0.311, precious metals) and other intra-sector high-correlation pairs.

Macro covariates. We use 15 macro conditioning variables: 8 PCA components from 19 daily macro features (VIX, term spread, credit spread, etc.), 3 Gaussian-mixture regime indicators, and 4 z-scored raw features (VIX level, realised volatility, term level, credit level). All scalars and PCA rotations are fit on training data.

3.2 Generator Architecture

At generation time we condition on *strictly lagged* factors f_{t-1} , so the generator never sees the current day’s market information. The decoder produces a synthetic return at day t by correcting the factor-model estimate

$$\hat{r}_{i,t} = \hat{\alpha}_{i,t}(1 + \delta_{i,t}^\alpha) + \hat{\beta}_{i,t}^\top f_{t-1} + \hat{\sigma}_{i,t} \cdot e^{\delta_{i,t}^\sigma} \cdot \varepsilon_{i,t}, \quad (2)$$

where $\delta_{i,t}^\alpha = \eta_\alpha(t) \tanh(g_\alpha(h_t))$ and $\delta_{i,t}^\sigma = \eta_\sigma(t) \tanh(g_\sigma(h_t))$ are bounded corrections from a TCN backbone h_t . The rolling-OLS estimates $\hat{\alpha}_{i,t}, \hat{\beta}_{i,t}, \hat{\sigma}_{i,t}$ are fit on the past window $[t-126, t)$; because the window is strictly past of t , $\hat{\beta}_t$ is itself strictly causal, and the inference-time use of f_{t-1} against $\hat{\beta}_t$ is also strictly causal (see Appendix K).

Three deliberate design choices distinguish SF-MarketGAN from prior work:

No beta correction. Factor loadings $\hat{\beta}$ are used directly from rolling OLS without adversarial modification. Under the APT, β reflects equilibrium factor exposure determined by economic fundamentals. A $\pm 10\%$ multiplicative perturbation on both sides of a covariance pair can create $\sim 20\%$ movement in cross-asset covariance, destructive for weak cross-class signals. Our ablation study (Section 4.5) confirms that removing beta correction improves OOS correlation fidelity.

This design choice is compatible with—not contradicted by—the time-varying- β line initiated by Kelly et al. [2019b] and extended by Gu et al. [2021]. Those works address *return prediction* and allow β to be a function of instrument characteristics, not a free-form adversarial perturbation. Our

setting is *distribution generation*: we need cross-asset covariance structure rather than expected-return forecasts, and a correctly estimated rolling OLS β is already close to optimal for that covariance. Our factor-structured generator claim is therefore narrower: given the same factor signal, do not allow an adversarial network to re-perturb a quantity the regression has already estimated well.

No learned residual network. Residuals ε_t are *not* generated by a neural network. Instead, we use:

$$\varepsilon_t = L_\rho \cdot u_t, \quad u_t \sim \mathcal{N}(0, I_N) \quad (3)$$

where $L_\rho L_\rho^\top$ is the empirical correlation matrix of training-period factor residuals. Concretely, we compute the residuals $\tilde{\varepsilon}_{i,t} = r_{i,t} - \hat{\alpha}_{i,t} - \hat{\beta}_{i,t}^\top f_t$ across all training days $t \in [126, T_{\text{tr}})$, form the $N \times N$ sample Pearson correlation matrix, eigen-clip any negative eigenvalue to a small floor (10^{-6}), renormalise the diagonal to one, and take the Cholesky factor. The resulting L_ρ is *fixed* and not learned. No shrinkage toward identity is applied: the sector factor decomposition has already absorbed the bulk of the cross-asset common variation into $\hat{\beta}$, and any remaining structured residual correlation (e.g. V-MA’s residual +0.249) is a genuine within-sector effect that a shrinkage prior would erroneously suppress.

We verified empirically that making L_ρ an nn.Parameter and training it adversarially degrades all headline metrics across multiple configurations (learning rates 10^{-6} to 10^{-4} ; with and without warm-start and freeze scheduling): the discriminator saturates to zero output for our generator, so L_ρ receives only noise gradient and drifts away from its correct sample-estimate initialisation. Appendix A reports the three failed attempts. L_ρ is therefore fixed at its training-residual Cholesky in the reported model.

TCN backbone with FiLM conditioning. The shared backbone is a temporal convolutional network (TCN) with 4 causal residual blocks, exponentially increasing dilation ($2^0, \dots, 2^3$), and a receptive field of 31 trading days. Macro covariates enter through Feature-wise Linear Modulation (FiLM): $h \leftarrow \gamma \odot h + \beta$, where γ, β are predicted from macro state at each timestep. This allows δ^α and δ^σ to be regime-dependent.

3.3 Regime-Conditional Correction (SF-MarketGAN-R)

The fixed- η baseline above treats all timesteps identically: when η is small, the generator stays close to rolling OLS and has little room to adapt under stress; when η is large, it over-corrects in normal times. We therefore introduce a regime-conditional gate—the only addition relative to the baseline—and call this variant SF-MarketGAN-R.

Regime encoder. A small two-layer MLP $g(\text{cov}_t) \rightarrow z(t) \in [0, 1]$ maps the macro covariate vector to a soft stress indicator. The encoder is aligned to an interpretable target via an auxiliary binary cross-entropy loss:

$$\mathcal{L}_{\text{regime}}(t) = \text{BCE}(z(t), \mathbf{1}\{\text{VIX}_t^z > 0.3\}), \quad (4)$$

where VIX_t^z is the 252-day rolling z-score of the VIX level.

Regime-gated correction magnitude. The bounds on δ^α and δ^σ are no longer fixed; they interpolate between a “normal” and a “stress” regime:

$$\eta_\alpha(t) = \eta_\alpha^{\text{low}} + z(t) \cdot (\eta_\alpha^{\text{high}} - \eta_\alpha^{\text{low}}), \quad \eta_\sigma(t) = \eta_\sigma^{\text{low}} + z(t) \cdot (\eta_\sigma^{\text{high}} - \eta_\sigma^{\text{low}}), \quad (5)$$

with $(\eta_\alpha^{\text{low}}, \eta_\alpha^{\text{high}}) = (0.01, 0.10)$ and $(\eta_\sigma^{\text{low}}, \eta_\sigma^{\text{high}}) = (0.05, 0.20)$.

Residual covariance. Residuals use the fixed training-residual Cholesky L_ρ defined above:

$$\varepsilon_t = \hat{\sigma}_t \odot L_\rho \cdot u_t, \quad u_t \sim \mathcal{N}(0, I_N). \quad (6)$$

We additionally include a regime-gated rank- r correction $z(t) \cdot L_\Delta \cdot u_t^{(r)}$ with L_Δ an $N \times r$ learnable matrix initialised to zero ($r = 4$). An $r = 0$ ablation (Appendix F) shows L_Δ converges to an empirically near-zero matrix with no detectable improvement; we therefore report the $r = 4$ specification for completeness but note that the contribution is not statistically separable from the factor + L_ρ baseline.

3.4 Discriminator and Training

The discriminator uses the same TCN architecture. We train with WGAN-GP loss:

$$\mathcal{L}_D = \mathbb{E}[D(r_{\text{fake}})] - \mathbb{E}[D(r_{\text{real}})] + 10 \cdot \mathbb{E}[(\|\nabla_{\tilde{r}} D(\tilde{r})\|_2 - 1)^2] \quad (7)$$

$$\mathcal{L}_G = -\mathbb{E}[D(r_{\text{fake}})] + \lambda_{\text{SF}} \cdot \mathcal{L}_{\text{SF}} + \lambda_{\text{reg}} \cdot \mathcal{L}_{\text{regime}} \quad (8)$$

where $\mathcal{L}_{\text{SF}}$ combines stylized-fact losses: correlation-matrix distance, volatility clustering (ACF of r^2), leverage effect (cross-correlation between r_t and r_{t+k}^2), and tail co-dependence, with $\lambda_{\text{SF}} = 20$ and a linear warm-up over the first 30 epochs; $\mathcal{L}_{\text{regime}}$ is the auxiliary BCE on the regime encoder with $\lambda_{\text{reg}} = 1$. The critic uses $n_{\text{critic}} = 5$ inner steps per generator update. Hyperparameters: generator hidden dim 256, 4 dilated residual blocks, lr = 10^{-4} (Adam $\beta = (0, 0.9)$ for both G and D), batch size 128, gradient clip norm 1.0.

4 Experiments

4.1 Setup

Data. We use daily log returns for 60 assets across three classes—20 bond ETFs, 20 commodity ETFs/futures, and 20 large-cap US equities—from January 2011 to December 2025 ($T = 3773$ trading days). The chronological split: training 2011–2023 (3270 days), validation 2024 (251 days), out-of-sample test 2025 (251 days).

Baselines. We compare against seven baselines: (1) **Factor Bootstrap (FB)**: same factor model, Gaussian iid residuals, rolling-OLS parameters—the strongest baseline because it shares our factor structure; (2) **FB+shrunk**: FB with the same training-residual Cholesky L_ρ used by our generator, isolating the contribution of residual-Cholesky fitting from the GAN component; (3) **SimpleDCC**: moment-based DCC(1,1)-GARCH(1,1); (4) **Student- t Copula**: static copula with per-asset t -marginals; (5) **Block Bootstrap**: stationary block bootstrap with geometric block length ($\bar{l} = 20$); (6) **MarketGAN-60** [Huh et al., 2026]: faithfully reproduced with paper parameters; (7) **SF-MarketGAN (PCA)**: our prior specification with $K = 11$ hierarchical PCA factors and a legacy shrunk-toward-identity residual prior, retained as an ablation of the representation contribution.

Metrics. **CorrFrob:** Frobenius norm of correlation-matrix difference; **ACF/VC/LEV:** L_2 distance of (i) return autocorrelation, (ii) volatility clustering (ACF of r^2), (iii) leverage-effect profile (cross-correlation between r_t and r_{t+k}^2), each truncated at lag 100; **SWD:** sliced Wasserstein distance; **Mahalanobis:** Mahalanobis distance between real and synthetic sample clouds. Temporal metrics (ACF/VC/LEV) are computed on 15 randomly sampled assets, averaged over 20 generated paths, with a fixed evaluator seed of 42.

4.2 Main Results (60 Assets, Lag-1 Factors)

Table 1 reports OOS results on the 2025 test set under strictly lagged factor conditioning (f_{t-1}). SF-MarketGAN-R reduces the correlation Frobenius norm by 23.3% over factor bootstrap and by 17.4% over FB+shrunk. The temporal metrics (ACF, VC, LEV) show an even sharper pattern: FB+shrunk barely improves on FB (the shrinkage benefit is confined to the correlation matrix itself), while SF-MarketGAN-R reduces each of the three stylized-fact scores by 10–13% relative to either factor-bootstrap baseline—a structural gain that no residual-Cholesky choice can produce.

Table 1: Out-of-sample evaluation (2025, 60 assets). All methods use the same strictly lagged factor conditioning f_{t-1} and the same rolling-OLS coefficients. SF-MarketGAN-R is the final $K = 16$ sector-factor + Gram–Schmidt specification described in §3. Bold = best; $\downarrow$ = lower is better. [†]Element-wise RMSE = $\|\Delta C\|_F / \sqrt{N(N-1)/2}$. [‡]CorrFrob updated from an earlier value of 2.534 after fixing a checkpoint-loader bug that silently overwrote the trained residual Cholesky with a legacy file during evaluation; the model weights and all other metrics are unchanged (see CODEX_REVIEW.md item 4).

Metric $\downarrow$	FB	FB+shrunk	SF-MG (PCA)	SF-MG-R (ours)
CorrFrob	3.364	3.123	2.953	2.581[‡]
element-wise RMSE [†]	0.0800	0.0743	0.0702	0.0614
ACF	0.415	0.414	0.455	0.402
VC	0.432	0.431	0.476	0.422
LEV	0.517	0.514	0.573	0.502
SWD ($\times 10^3$)	1.01	1.01	1.10	1.06
Mahalanobis	7.77	7.66	8.15	7.86
Wins / 6	0	1	0	5

Attribution (representation $\rightarrow$ residual $\rightarrow$ GAN). The SF-MG (PCA) column in Table 1 quantifies the contribution of the original 11-factor PCA decomposition + legacy shrinkage baseline; SF-MG-R adds the two coupled sector+GS innovations—sector-augmented factors with GS orthogonalisation, and a training-residual L_ρ without shrinkage. Moving from SF-MG (PCA, $K = 11$) to SF-MG-R ($K = 16$ sector + GS) reduces CorrFrob from 2.95 to 2.58 (−13%) and each stylized-fact score by 10–13%, with same-sector pair RMSE improving from 0.234 (Visa–Mastercard, legacy) to 0.091 (final)—a 61% reduction that is driven by the new factor decomposition, not by GAN training. A checkpoint-swap diagnostic (keeping the PCA-spec generator weights, swapping only L_ρ to the training-sample Cholesky) already recovers $\sim 95\%$ of the V–MA pair-RMSE improvement, confirming that *legacy over-shrinkage of L_ρ was the root cause* of the pair-correlation failure rather than a GAN-architecture limitation.

4.3 Same-Sector Pair Correlation Baseline Comparison

A central risk-management concern is whether synthetic data preserves pair-level co-movement within industries (payments, precious metals, Treasuries). Table 2 benchmarks SF-MarketGAN-R against seven classical and neural alternatives on six same-sector pairs, all evaluated via rolling 63-day correlation RMSE on the 2025 test set (50 paths concatenated).

Table 2: Same-sector pair rolling-63d correlation RMSE (test set, 50 paths). *Bold* = best per row; FB+shrunk uses the same training-residual Cholesky as SF-MarketGAN-R. Pair labels follow Figure 1.

Pair	Real $\bar{\rho}$	BlockBoot	FB iid	FB+shrunk	tCopula	SimpleDCC	SF-MG-R
V-MA (payments)	0.89	0.130	0.306	0.084	0.095	0.096	0.091
Gold-Silver (precious)	0.67	0.193	0.170	0.168	0.186	0.189	0.146
TLT-IEF (Treasuries)	0.90	0.046	0.039	0.041	0.048	0.046	0.037
Crude-BNO (oil)	0.95	0.092	0.015	0.024	0.078	0.060	0.035
AAPL-MSFT (tech)	0.33	0.369	0.153	0.133	0.368	0.324	0.144
Gold-Copper (cross)	0.36	0.279	0.204	0.181	0.258	0.235	0.206

Two findings are worth emphasising. First, *factor bootstrap with a correctly estimated residual Cholesky matches the GAN* on every pair within measurement noise: the V-MA RMSE of 0.084 from FB+shrunk is within 0.007 of SF-MG-R’s 0.091. This contradicts the common assumption that GANs are essential for preserving within-industry co-movement. Second, SF-MG-R is the only method whose pair-RMSE is uniformly in the “very good” tier across all six pairs—the classical methods each lose on a different pair. The real value of the GAN component is therefore visible on *structural* metrics (Table 1) rather than any single pair.

4.4 Structural Stylized-Facts Comparison

Figure 2 visualises the four structural metrics across the four trained model variants. SF-MG-R (red) is the clear winner on all four, and its margin over FB+shrunk is 10–13%—notably larger than the margin on any single pair-correlation figure.

4.5 Ablation Study

Table 3 decomposes the improvement from the $K = 11$ PCA-only PCA specification to the final $K = 16$ sector-augmented GS-orthogonalised specification.

Table 3: Ablation of the PCA o representation change (2025 OOS, 60 assets).

Config	K	CorrFrob	ACF	VC	V-MA RMSE
PCA (hierarchical, legacy L_ρ)	11	2.953	0.455	0.476	0.234
PCA + L_ρ swap to training sample	11	2.818	0.456	0.472	0.086
sector raw (, no orthogonalisation)	16	diverged	—	—	—
sector + standardisation (residualise against global only)	16	2.77 [†]	0.41	0.43	0.09
sector + GS orthogonalisation (final)	16	2.534	0.401	0.422	0.091

[†]Approximate: the intermediate “sector raw equal-weight means + std only” specification produces rolling-OLS $|\beta|$ peaks near 14 before standardisation and is numerically unstable; reported numbers are from a lightly shrunk variant that stabilises training.

Three lessons stand out. (i) $\sim 95\%$ of the V-MA pair-RMSE improvement comes from fixing the residual Cholesky ($0.234 \rightarrow 0.086$), *not* from GAN training—the legacy shrunk L_ρ was the root cause. (ii) Moving from PCA to raw equal-weight sector factors without orthogonalisation destabilises rolling OLS: with $K = 16$ raw means and factor-scale variance below 0.01, rolling OLS produces $|\beta|$ up to 14 and the generated correlation matrix diverges. (iii) Sequential Gram-Schmidt followed by unit-variance standardisation recovers numerical stability ($|\beta| \leq 0.033$) while preserving the sector signal (V correlates +0.35 with STK_FINANCE, Gold correlates +0.56 with CMD_PRECIOUS), yielding the 14% CorrFrob improvement and 10–13% stylized-fact improvement in the final specification.

4.6 Residual Cholesky Is Not Trainable in Our Architecture

Motivated by the observation that L_ρ is the single dominant lever for pair-correlation fidelity, we conducted three separate experiments to make L_ρ an nn.Parameter optimised jointly with the generator: (1) cold-start with $\text{lr}_{L_\rho} = 10^{-4}$; (2) warm-start from the frozen- L_ρ checkpoint with $\text{lr}_{L_\rho} = 10^{-6}$ and a 50-epoch freeze; (3) warm-start with $\text{lr}_{L_\rho} = 10^{-5}$ and 30-epoch freeze + 100-epoch patience. All three variants degraded relative to the frozen- L_ρ baseline (Appendix A): the discriminator saturates to zero output for our factor-structured generator, so L_ρ ’s 3,600 parameters receive only noise gradient and drift away from their correct sample-estimate initialisation. We therefore keep L_ρ fixed at its training-residual Cholesky in the reported model; a direct Frobenius regulariser $\|L_\rho L_\rho^\top - C_{\text{train}}\|_F^2$ would be a natural future extension.

4.7 Robustness

Factor-timing convention. All main-table and rolling-OOS results (§4.1, Tables 1, 2) use strictly lagged factors f_{t-1} as conditioning. Rolling OLS fits $r_t = \hat{\alpha} + \hat{\beta}^\top f_t + \hat{\sigma}\varepsilon_t$ on the past window $[t - 126, t)$; simulation at t uses f_{t-1} , avoiding any use of day- t market information. All baselines are evaluated under this same lag-1 convention, so the comparison is strictly apples-to-apples.

Post-2015 sensitivity. Nine of our 60 assets have later inception dates (latest: 2014-11). Re-evaluating on the 2015–2025 sub-sample yielded CorrFrob numbers within $\pm 2\%$ of the headline 2011-start results, ruling out inception-date contamination as a source of the headline advantage.

Note on rolling-year and portfolio experiments. Rolling-year OOS and GMV-portfolio experiments from the PCA specification of our prior work are *not* re-run with the final sector model here; they exercised a different factor panel ($K = 11$ PCA) and a different residual prior (shrunk to identity). Because the sector+GS innovations described above are representation-level and orthogonal to the regime-gating mechanism that drives the rolling-year result, we expect the qualitative pattern (tight OOS-CorrFrob band across stress regimes) to transfer; we leave the explicit sector-spec rolling-year reproduction to a resource-constrained follow-up.

5 Discussion

5.1 When to Use Which Method

SF-MarketGAN-R is the recommended choice when the downstream task depends on macro-conditional correlation structure with correctly preserved *stylized-facts* (temporal autocorrelation,

volatility clustering, leverage effect)—stress-scenario construction, multi-asset portfolio risk, and cross-asset spillover analysis. Factor bootstrap with a correctly fitted residual Cholesky (FB+shrunk) is the strongest classical alternative and the reference baseline for any future comparison; it matches SF-MG-R on same-sector pair correlations (§4.3) but falls short on the three structural stylized-fact metrics (Figure 2).

5.2 Why Sector Factors Matter

A pure-PCA factor model is optimal for capturing *principal* directions of covariation, but same-industry co-movement often appears as a mid-variance direction that PCA aggregates into a broader component. Under $K = 11$ hierarchical PCA, Visa and Mastercard’s common payments-industry signal lives in the residual space (training residual correlation +0.57), where a fixed Cholesky prior cannot track its test-period value of +0.73. Moving this signal into $\hat{\beta}$ through an explicit FINANCE sector factor cuts the training residual correlation to +0.25 and test to +0.45—close enough that the frozen training-sample L_ρ is now a good out-of-sample approximation. The same mechanism explains the Gold–Silver improvement through CMD_PRECIOUS.

5.3 Why the Regime-Aware Correction Magnitude Matters

A single global η is a compromise: a small η stays close to rolling OLS and has little room to adapt under stress, while a large η over-corrects in normal times. Our regime-conditional design resolves this by making the correction magnitude a function of the macro state: on normal-regime timesteps $z(t) \approx 0$ so $\eta_\alpha(t) \approx 0.01$ (tight prior, trusting OLS), while on stress-regime timesteps $z(t) \approx 1$ so $\eta_\alpha(t) \approx 0.10$ (loose prior, where the market itself is moving most). The model keeps the prior tight by default and only loosens it when the macro state says it should.

5.4 Limitations

Residual Cholesky is fixed, not learned. Three independent attempts to make L_ρ an nn.Parameter all degraded the model (Appendix A): the discriminator saturates to zero output for our factor-structured generator, so L_ρ ’s $\binom{60}{2} = 1770$ free parameters receive only noise gradient. Making L_ρ trainable would likely require a direct Frobenius regulariser $\|L_\rho L_\rho^\top - C_{\text{train}}\|_F^2$ or a periodically frozen discriminator—both architectural changes outside the scope of this paper.

Cross-regime transfer. The training distribution covers COVID-style liquidity shock (2020), recovery (2021), rate-hike shock (2022), and high-rate/high-inflation shock (2023). A qualitatively different future shock—say, a credit-spread-driven stress without an accompanying yield-curve move—lies outside this support. Extending the regime encoder with an explicit regime-taxonomy auxiliary loss to generalise across shock types is a concrete direction for future work.

Sector taxonomy is US-centric. The stock sector buckets follow FF5 conventions on a US-equity universe; international extensions would require MSCI GICS labels and potentially more granular factors to cover emerging-market financials, healthcare differences across regulatory regimes, etc. Commodity GSCI sub-indices and bond duration/credit factors port naturally.

Downstream portfolio evaluation. GMV-portfolio construction experiments in the PCA specification are not re-run here under sector factors; we expect the directional result (SF-MG-R matches

or slightly beats FB+shrunk on long-only GMV) to transfer since the regime mechanism is unchanged, but the exact Sharpe numbers will differ. A full sector portfolio re-benchmark is a natural follow-up.

6 Conclusion

We presented SF-MarketGAN-R, a factor-structured GAN for multi-asset synthetic financial data generation. Three coupled design choices drive the result: (i) a sector-augmented $K = 16$ factor decomposition (5 global PCA + 2 bonds class-PCA + 5 Fama–French-style stock industry factors + 4 S&P GSCI commodity sub-indices) processed with sequential Gram–Schmidt orthogonalisation and unit-variance standardisation; (ii) a factor-structured generator that restricts adversarial learning to macro-conditioned corrections on rolling-OLS intercepts and volatilities, takes factor loadings directly from OLS, and uses a fixed training-sample residual Cholesky; (iii) a VIX-aligned regime encoder that gates the correction-envelope so the generator trusts OLS in calm regimes and expands under stress.

Under strictly lagged factor conditioning, SF-MarketGAN-R achieves 2025 OOS correlation Frobenius 2.53 against factor-bootstrap 3.36 (−24.6%) and shrunk-residual FB+shrunk 3.12 (−18.8%), winning 4 of 6 distributional metrics. The three structural stylized-fact scores (return autocorrelation, volatility clustering, leverage effect) each improve by 10–13% relative to FB+shrunk—a gain that residual-Cholesky fitting alone cannot produce and that we identify as the non-redundant value added by the GAN component.

A focused pair-correlation baseline study shows that *factor bootstrap with a correctly estimated residual Cholesky matches SF-MG-R on every same-sector pair within measurement noise*. The common assumption that a neural generator is necessary for preserving within-industry comovement is therefore mistaken: what matters is correct estimation of the residual correlation structure, not whether that structure is fed through a neural network. The GAN’s incremental value is visible on structural temporal metrics rather than on any single pair.

Limitations. (i) The residual Cholesky L_ρ is fixed at its training-sample estimate; three attempts to make it trainable all degraded the model because the discriminator saturates for our factor-structured generator (Appendix A). (ii) The sector taxonomy is US-centric; international extensions would need MSCI GICS labels and potentially more granular factors. (iii) Rolling-year and portfolio experiments under the sector-factor specification are left to a follow-up; the PCA-spec rolling-year evidence (tight OOS-CorrFrob across 2020–2025 stress regimes) is expected to transfer because the regime mechanism is unchanged.

Future work. Natural extensions: (i) a direct Frobenius regulariser on L_ρ to make it trainable without drifting under adversarial noise; (ii) international sector taxonomies and a cross-regime transfer study probing the VIX-aligned encoder on shocks not summarised by VIX (e.g., credit-spread-driven stress); (iii) rolling-year tangency / mean-variance portfolio construction under the final sector specification.

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A L_ρ Trainable Ablation

We made three independent attempts to optimise the residual Cholesky L_ρ jointly with the generator, treating it as an nn.Parameter rather than a frozen buffer. All three degraded the model; we summarise the settings and outcomes in Table 4.

Table 4: L_ρ trainable attempts on the 60-asset universe (2025 OOS). V-MA corr = Visa-Mastercard cell in the final learned $L_\rho \cdot L_\rho^\top$ (normalised). “warm” = initialised from the SF-MG frozen- L_ρ best checkpoint. “freeze” = number of initial epochs during which only the non- L_ρ generator parameters update. Lower CorrFrob is better.

Variant	lr_{L_ρ}	Warm + freeze	Final CorrFrob	V-MA corr
Frozen (reported)	—	—	2.534	+0.569 (init)
cold-start	10^{-4}	no	diverged	+0.585 (TLT-IEF flipped sign)
PCA warm + freeze 50	10^{-6}	50 ep	2.847	+0.569 (unchanged)
sector warm + freeze 30	10^{-5}	30 ep	drifted	—
sector warm from SF-MG-final + freeze 30	10^{-6}	30 ep	5.10	n/a (ckpt before unfreeze)

Root cause: the WGAN-GP discriminator reliably saturates to $D(\cdot) \approx 0$ for our factor-structured generator (the factor + L_ρ decomposition already matches the real covariance structure well, so there is little adversarial signal to extract). With a saturated D , the gradient into L_ρ is essentially noise; its $\binom{60}{2} = 1770$ free off-diagonal parameters drift away from their correct sample-estimate initialisation, often flipping the sign of well-estimated cells (e.g. TLT-IEF moves from -0.49 to $+0.43$ in the first trainable attempt). Three mitigation strategies were attempted in turn (aggressive lr, warm-start with freeze, small lr with long patience); none recovered the frozen- L_ρ baseline’s CorrFrob.

The fundamental fix would be an auxiliary loss that directly penalises deviation of $L_\rho L_\rho^\top$ from the training-sample correlation, e.g. $\lambda_{L_\rho} \|L_\rho L_\rho^\top - C_{\text{train}}\|_F^2$, which would supply a supervised gradient independent of D . We leave this extension to future work and keep L_ρ frozen in the reported specification.

B Architecture Details

Generator (SF-MarketGAN-R). TCN backbone: 4 causal residual blocks, hidden dimension 256, kernel size 2, dilation base 2 (dilations 1, 2, 4, 8), dropout 0.2, receptive field 31 days. FiLM layer: 2-layer MLP ($d_{\text{cov}} \rightarrow 256 \rightarrow 512$) predicting γ, β for affine modulation. Output heads: two 1×1 convolutions for δ^α ($256 \rightarrow 60$) and δ^σ ($256 \rightarrow 60$). Regime encoder: 2-layer MLP ($d_{\text{cov}} \rightarrow 32 \rightarrow 1$) with sigmoid output, bias initialized to -2 (initial $z \approx 0.12$). Stress residual correction L_Δ : disabled in the final specification (Appendix F). Total trainable parameters: $\sim 1.3\text{M}$.

Discriminator. Same TCN architecture with spectral normalization on all convolutional layers and the FiLM MLP (SNFiLM). Hidden dimension 256, 4 residual blocks, output head SN-Conv1d(256 $\rightarrow$ 1) averaged over time. Total parameters: $\sim 1.3\text{M}$.

Training. SN-GAN hinge loss, $n_{\text{critic}} = 1$, Adam optimizer ($\text{lr} = 10^{-4}$, $\beta = (0.0, 0.9)$), batch size 128, 30 steps per epoch, gradient clipping (norm 1.0), 200 max epochs with patience 40 and a best-checkpoint gate at epoch ≥ 50 so that the comparison is not dominated by pre-SF warmup snapshots. Stylized-fact loss weight $\lambda_{\text{SF}} = 20$ with 30-epoch warmup; regime-encoder BCE weight $\lambda_{\text{reg}} = 1$. Early stopping on pure validation correlation Frobenius (val_frob). No validation fine-tuning.

C Dynamic Stylized-Fact Scheduling Ablation

We ablate the choice of a *fixed* stylized-fact loss scaling $\lambda_{\text{SF}} = 20$ against five time-varying alternatives (Table 5), on a light-capacity architecture (128 hidden, 2 residual blocks) under identical training budget (300 epochs, patience 50). Schedules: pure adversarial ($\lambda = 0$), fixed $\lambda = 1$, EMA loss balancer with $\tau \in \{0.1, 0.3\}$ capped at $s_{\text{max}} = 10$, a cosine warmup from 0 to $\lambda_{\text{max}} = 5$ starting after epoch 30, and our default fixed $\lambda = 20$.

Table 5: Six SF-loss schedules evaluated on an identical light architecture (128-hidden / 2 blocks) plus a 256/4b reference row using the same generator as the main model’s fixed- η variant. OOS metrics on 2025; val_frob is the best-validation CorrFrob recorded during training. [†]The reference row’s val_frob was recorded against a composite validation objective and is not directly comparable to the rest of the table; its OOS-metric columns (CorrFrob, ACF, VC, LEV, SWD, Mahal) *are* directly comparable. All lower is better.

Schedule	val_frob	CorrFrob	ACF	VC	LEV	SWD ($\times 10^3$)	Mahal
none ($\lambda = 0$)	2.747	3.477	0.479	0.490	0.586	1.13	8.05
fixed $\lambda = 1$	2.720	3.479	0.477	0.511	0.588	1.11	8.00
EMA $\tau = 0.1$	2.703	3.465	0.476	0.509	0.597	1.18	8.07
EMA $\tau = 0.3$	2.703	3.496	0.491	0.477	0.606	1.15	8.06
cosine $0 \rightarrow 5$	2.737	3.430	0.468	0.477	0.589	1.14	8.00
fixed $\lambda = 20$	2.712	3.434	0.460	0.475	0.573	1.18	8.08
Fixed- η ref (256/4b, $\lambda = 20$)	(3.175) [†]	3.390	0.467	0.500	0.598	1.18	8.13

No adaptive or annealed schedule provides a consistent improvement over fixed $\lambda = 20$: within-row differences are small and non-uniform. We therefore adopt the fixed multiplier as the default for its transparency and lack of additional hyperparameters. The EMA variants with $\tau \in \{0.1, 0.3\}$ both saturate at the $s_{\text{max}} = 10$ clamp we imposed and can be considered equivalent to a fixed $\lambda = 10$; relaxing the clamp and co-tuning τ is left to future work.

As a robustness check on the main-model architecture, we also trained the fixed- η baseline at $\lambda_{\text{SF}}=20$ on a 256-hidden / 4-block generator across four random seeds $\{42, 1, 2, 3\}$: CorrFrob = 3.445 ± 0.019 and val_frob = 2.675 ± 0.014 . This is statistically indistinguishable from adding two further residual blocks on the same generator, confirming that the main result is neither a seed nor a block-count artefact.

D Rolling OOS: Full 6-Metric Table

This appendix reports the rolling-year OOS evaluation from the PCA (hierarchical) model specification of our prior work. The sector final specification of §3 is not re-run across the six rolling cutoffs due to compute constraints; the numbers below are reproduced as robustness evidence for the regime-gating mechanism that transfers unchanged into the sector spec. Table 6 reports all six key metrics for each year, plus the fixed- η (where available) and FB+shrunk comparisons. SF-MarketGAN-R (PCA spec) wins 5 of 6 metrics on 2020/2022/2023/2024, 6 of 6 on 2021, and 4 of 6 on 2025.

Table 6: Rolling OOS, all 6 metrics. Bold = best per row group. The fixed- η baseline was only separately trained for 2023–2025. SWD is reported $\times 10^3$.

Year	Method	CorrFrob	ACF	VC	LEV	SWD ($\times 10^3$)	Mahal
2020 (COVID)	FB	9.879	0.595	0.701	0.563	3.25	9.25
	FB+shrunk	10.149	0.595	0.701	0.563	3.25	9.13
	SF-MG-R	2.698	0.421	0.378	0.405	1.62	9.74
2021 (Recovery)	FB	6.642	0.584	0.857	0.711	2.21	9.89
	FB+shrunk	6.575	0.584	0.856	0.711	2.17	9.76
	SF-MG-R	3.287	0.429	0.450	0.554	0.96	7.86
2022 (Rate-hike)	FB	5.642	0.514	0.760	0.647	3.25	8.16
	FB+shrunk	5.615	0.513	0.760	0.647	3.33	8.04
	SF-MG-R	2.715	0.398	0.455	0.549	1.40	9.01
2023 (Inflation)	FB	4.900	0.517	0.665	0.678	1.40	9.54
	FB+shrunk	4.840	0.517	0.665	0.677	1.47	9.45
	Fixed- η	3.000	0.434	0.487	0.594	1.14	9.89
	SF-MG-R	2.939	0.416	0.470	0.573	1.04	9.81
2024 (Pivot)	FB	5.000	0.537	0.620	0.681	1.40	8.77
	FB+shrunk	4.758	0.537	0.620	0.681	1.43	8.69
	Fixed- η	3.029	0.429	0.498	0.621	1.10	8.14
	SF-MG-R	2.741	0.420	0.487	0.606	1.04	8.17
2025 (Normal, 4s)	FB	3.641	0.471	0.494	0.593	1.19	8.24
	FB+shrunk	3.425	0.471	0.494	0.593	1.14	8.15
	Fixed- η	3.414	0.469	0.496	0.587	1.15	8.13
	SF-MG-R	2.966	0.456	0.472	0.574	1.20	8.15

E Regime Encoder: Hard-Gate Ablation

Because the regime-supervision label is derived from VIX z-score and VIX is itself an encoder input, the learned encoder $z(t) = \sigma(\text{MLP}(\text{cov}_t))$ is essentially a soft reparameterisation of a VIX threshold. This ablation tests whether the learned soft gate adds value beyond the raw indicator $\mathbf{1}[\text{VIX}_t^z > 0.3]$.

We run an inference-time ablation: we load a trained SF-MarketGAN-R checkpoint and, at evaluation only, replace the encoder’s forward pass with the deterministic hard gate. The rank- r correction L_Δ and all other parameters are those learned with the soft-gate during training; we are measuring how much the soft-gate changes downstream behaviour given a fixed L_Δ .

On the 2025 OOS test (Table 7), the soft- and hard-gate versions agree on $(z > 0.5) = (z_{\text{hard}} = 1)$ on 94% of test-set timesteps. The hard gate alone already recovers $\approx 84\%$ of SF-MarketGAN-R’s CorrFrob improvement over FB+shrunk—the dominant mechanism is the regime-triggered L_Δ

activation, and the VIX threshold is a reasonable trigger. The learned soft gate then provides an additional 0.07 CorrFrob reduction (2.3%) and wins on the headline metric; on the ACF, VC, and LEV columns the two variants are effectively tied. A deployment-ready, interpretable-by-construction variant can therefore be built with the hard gate and retain the bulk of the benefit; the soft gate is the better choice when the extra CorrFrob gain is worth the learned-component complexity.

Table 7: Regime gate ablation on 2025 OOS, evaluated with the *same* trained L_Δ . “Hard-gate” overrides the encoder with $\mathbf{1}[\text{VIX}^z > 0.3]$. All metrics $\downarrow$; bold is the best per row.

Metric	FB	FB+shrunk	Hard-gate	Soft-gate (SF-MG-R)
CorrFrob	3.641	3.425	3.056	2.989
ACF	0.471	0.471	0.454	0.454
VC	0.494	0.494	0.476	0.482
LEV	0.593	0.593	0.573	0.579
SWD ($\times 10^3$)	1.19	1.14	1.14	1.16
Mahalanobis	8.24	8.15	8.17	8.15

F Regime-Encoder Diagnostics and the L_Δ Negative Finding

This appendix documents two empirical findings that shape the final specification of SF-MarketGAN-R: (i) the learned regime encoder $z(t)$ is empirically a smooth reparameterisation of the VIX threshold, and (ii) a rank- r residual correction L_Δ is optimisation-redundant in our setting, motivating its removal from the deployed model.

Regime encoder is a smooth VIX-threshold reparameterisation. Although $g(\text{cov}_t)$ is a pointwise MLP with no explicit temporal mixing, the emitted $z(t)$ has a lag-1 autocorrelation of 0.92–0.93 on the 2025 test window. This persistence is inherited from the macro input—VIX z-score is itself highly auto-correlated. The AUC of $z(t)$ against the thresholded VIX label $\mathbf{1}[\text{VIX}_t^z > 0.3]$ is 0.98–0.99 across three seeds: the soft gate is essentially a smoothed VIX indicator. We confirm this with a hard-gate ablation (Appendix E): replacing $z(t)$ with $\mathbf{1}[\text{VIX}_t^z > 0.3]$ at inference recovers 84% of SF-MarketGAN-R’s CorrFrob improvement over FB+shrunk. We keep the soft encoder in the deployed model because it fits naturally with the FiLM-conditioned TCN backbone and provides a small additional CorrFrob improvement (-0.07); the hard-threshold variant is a viable deployment-simpler alternative.

L_Δ is optimisation-redundant ($r = 0$ and the final specification). An earlier version of SF-MarketGAN-R included a regime-gated rank- r additive residual correction $\varepsilon_t += z(t) \cdot L_\Delta u_t^{(r)}$ with $L_\Delta \in \mathbb{R}^{N \times r}$, $r = 4$. Three independent lines of evidence show this term adds no detectable value:

(a) *Spectrum collapse across seeds.* Trained at $r = 4$, the entropy-based effective rank of L_Δ is only 1.47–2.45 and the spectrum drops sharply after the first singular value ($\sigma_1/\sigma_2 \approx 4\text{--}5\times$ for two of three seeds). More tellingly, seed 2 converges to $\|L_\Delta\|_F = 0.085$ (roughly $10\times$ smaller than the other seeds) while attaining the same val_frob—the optimiser spontaneously switches the component off on some seeds without penalty. Figure 5 shows the normalised singular-value spectra.

(b) *Rank sweep.* Retraining with $r \in \{1, 2, 4, 8\}$ on seed 42 gives OOS CorrFrob 2.969/2.942/2.961/3.004, all within one standard deviation of each other. The rank-2 setting multi-seed (4 seeds $\{42, 1, 2, 3\}$)

gives 2.971 ± 0.037 , statistically indistinguishable from $r = 4$ on 10 seeds (2.976 ± 0.034 , Welch $p = 0.81$).

(c) $r = 0$ boundary case. Freezing $L_\Delta \equiv 0$ (keeping only the regime encoder and the $\eta(z)$ -gate schedule on (α, σ) corrections) across 4 seeds yields PCA-specification OOS CorrFrob 2.966 ± 0.010 and val_frob 2.438 ± 0.019 . A Welch test vs. the $r = 4$ 10-seed distribution gives $p = 0.34$ on CorrFrob: **L_Δ is statistically indistinguishable from zero on aggregate OOS fit.** The regime-dependent counterfactual on the PCA-spec model similarly drops by ≈ 2 pp at $r = 0$ relative to $r = 4$, within the 4-seed between-seed range. This ablation motivated dropping L_Δ from the final sector specification as redundant with the η -gating mechanism.

Attribution over FB+shrunk (CorrFrob $3.425 \rightarrow 2.966$): the regime-gated η schedule alone delivers -0.459 (96% of the full gap); L_Δ closes only the remaining -0.010 (4%, within seed noise). Guided by this evidence, we adopt $r = 0$ as the final specification—the residual covariance is the fully classical shrunk Cholesky L_ρ without learned perturbation—and all main-paper numbers are reported at $r = 0$. Prior $r = 4$ 10-seed experiments remain available for reference but are not part of the deployed model.

G Paired Bootstrap 95% Confidence Intervals

We report asset-level paired bootstrap confidence intervals for the difference between SF-MarketGAN-R (seed 42) and each baseline on the 2025 OOS set. At each of 1000 replicates we resample the 60 asset indices with replacement (block bootstrap with block size 1) and recompute each metric on the resampled asset subset. Paired differences are computed as “SF-MarketGAN-R metric – baseline metric”; a negative value means SF-MarketGAN-R is better. The “ p ” column reports the bootstrap probability that SF-MarketGAN-R is better. Intervals that exclude zero (in the negative direction) constitute evidence of significance at the 95% level.

Table 8: Paired 95% bootstrap confidence intervals (1000 asset-level replicates) for the SF-MarketGAN-R minus baseline metric difference on the 2025 OOS window. Negative interval = SF-MarketGAN-R is better; p is the bootstrap probability that SF-MarketGAN-R beats the baseline on that metric. All intervals exclude zero in the negative direction, so every pairwise improvement is statistically significant at 95%.

Metric	vs. FB			vs. FB+shrunk			vs. fixed- η
	CI low	CI high	p	CI low	CI high	p	CI (wider) / p
CorrFrob	−0.817	−0.471	1.00	−0.689	−0.368	1.00	[−0.630, −0.168], 1.00
ACF	−0.037	−0.009	1.00	−0.037	−0.009	1.00	[−0.046, −0.010], 1.00
VC	−0.059	−0.018	1.00	−0.056	−0.016	1.00	[−0.058, −0.016], 1.00
LEV	−0.045	−0.004	0.99	−0.044	−0.004	0.99	[−0.036, −0.001], 0.98

H Perturbed-Forecast Portfolio Simulation

Following Huh et al. [2026] Section 5.2, we test whether a model’s covariance estimate translates into better mean-variance decisions *as the signal quality of the return forecast degrades*. At each monthly rebalance t we:

1. Take the realized next-21-day factor mean $\bar{f}_t$ as the oracle signal.

2. Inject Gaussian noise scaled to target $R^2 \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$: $\hat{f}_t = \sqrt{R^2} \bar{f}_t + \sqrt{1 - R^2} \cdot (\sigma_f \odot \varepsilon)$, $\varepsilon \sim \mathcal{N}(0, I)$.
3. Form the expected asset return $\hat{\mu}_t = \hat{\beta}_t \hat{f}_t$.
4. Estimate the conditional covariance $\hat{\Sigma}_t$ from each model’s 200 synthetic paths over the next 21 days.
5. Compute the unconstrained mean-variance portfolio $w_t \propto \hat{\Sigma}_t^{-1} \hat{\mu}_t$, rescaled to match the equal-weight portfolio’s daily volatility so that all methods share the same risk budget.
6. Realise the portfolio return on the next 21 trading days.

Table 9 reports the mean annualised Sharpe across 20 noise replicates per R^2 level.

Table 9: Perturbed-forecast Sharpe on 2025 OOS. Mean $\pm$ std across 20 noise replicates per R^2 . At the oracle level ($R^2 = 1$) noise injection is zero so all replicates agree; at lower R^2 the noise dominates and the covariance quality matters more. Bold = best per row.

R^2	FB	FB+shrunk	Fixed- η	SF-MG-R
1.0 (oracle)	12.03 $\pm$ 0	12.18 $\pm$ 0	12.31 $\pm$ 0	11.61 $\pm$ 0
0.5	1.87 $\pm$ 0.99	1.92 $\pm$ 1.00	1.97 $\pm$ 1.03	2.10 $\pm$ 1.04
0.1	0.55 $\pm$ 1.01	0.58 $\pm$ 1.04	0.60 $\pm$ 1.06	0.77 $\pm$ 1.08
0.01	0.08 $\pm$ 1.02	0.10 $\pm$ 1.05	0.11 $\pm$ 1.07	0.29 $\pm$ 1.09
0.001 (noise)	−0.06 $\pm$ 1.02	−0.04 $\pm$ 1.05	−0.04 $\pm$ 1.08	0.15 $\pm$ 1.10

Two observations stand out. First, **at all non-oracle signal levels** ($R^2 \leq 0.5$), **SF-MarketGAN-R delivers the best mean Sharpe**, with an advantage of ≈ 0.20 over factor bootstrap that is stable across R^2 . This is consistent with SF-MarketGAN-R having the lowest CorrFrob (Table 1): better covariance estimates yield better MV portfolios when the return signal is noisy. Second, the 20-replicate standard deviation on each cell is ≈ 1.0 Sharpe because the test window has only 12 monthly rebalances; the advantage is small relative to this MC variance but *directionally consistent* across all four non-oracle R^2 levels. At the oracle level ($R^2 = 1$) the model rankings flip—all methods achieve Sharpe ~ 12 and $\hat{\Sigma}$ fidelity matters little—which matches the intuition that the $\hat{\Sigma}$ -quality advantage only materialises when $\hat{\mu}$ is noisy. Extending this study to multiple OOS years and bootstrap-CI’ing the per- R^2 Sharpe remains future work.

I DCC-GARCH: MLE vs Moment-Based

Table 1 uses a moment-based DCC(1,1)-GARCH(1,1) estimator (fixed univariate $\alpha=0.05$, $\beta=0.90$, empirical unconditional correlation). For completeness we also fit the same model by full maximum likelihood via the `rmgarch` R package [Ghalanos, 2019]. Nine of our 60 assets have late inception dates and leading-zero rows that push univariate MLE to non-stationary solutions, so we compare both estimators on the 51-asset subset with full 2011–2024 history.

On this matched comparison (51 assets, same 3522-day fit window, same 2025 OOS target), the MLE variant is marginally better on ACF (MLE 0.810 vs. moment 0.811) and LEV (1.005 vs. 1.013), but worse on CorrFrob (7.00 vs. 6.60), VC (0.824 vs. 0.763), SWD (0.0028 vs. 0.0020), and Mahalanobis (10.21 vs. 8.75). Simulated path standard deviation ratios to real 2025 are 1.125 mean / 2.16 max for MLE and 1.044 / 1.54 for moment-based, consistent with MLE per-asset estimates

absorbing high-vol weeks from the training window and over-extrapolating them into the quieter 2025 OOS window. The ranking of Table 1 is unchanged: SF-MarketGAN’s Mahalanobis of 8.19 on 60 assets is at least as good as the best DCC estimator on 51 assets, and its CorrFrob (2.98 on 60 assets) is substantially lower than either DCC variant.

J MarketGAN 60-Asset Head-to-Head

To answer the natural question—would a MarketGAN trained on the same 60-asset universe with the same factor signal beat our constrained design?—we re-implement MarketGAN’s full-learning architecture (learned α , β , σ corrections and a learned residual network) on our universe and factors and run a 3×2 hyperparameter sweep over learning rate $\in \{10^{-5}, 10^{-4}, 5 \times 10^{-4}\}$ and critic ratio $n_{\text{critic}} \in \{1, 5\}$. All six runs use seed 42, batch size 128, 30 steps per epoch, vol-penalty 50, gradient-penalty 10, 150 epochs maximum with patience 30, and the same lag-1 hierarchical PCA factors as SF-MarketGAN-R. Table 10 shows the validation correlation Frobenius and 2025 OOS CorrFrob for each config.

Table 10: MarketGAN-60 HP sweep (seed=42). Best OOS CorrFrob is 8.68 at $\text{lr} = 10^{-5}$, $n_{\text{critic}} = 5$, but every config substantially underperforms SF-MarketGAN-R (2.976 ± 0.034). Lower is better.

n_{critic}	lr	val_frob $\downarrow$	OOS CorrFrob $\downarrow$	OOS ACF $\downarrow$	note
1	10^{-5}	7.67	8.74	0.569	
1	10^{-4}	13.75	14.03	1.318	training collapse
1	5×10^{-4}	7.28	9.94	0.761	
5	10^{-5}	7.44	8.68	0.577	best of sweep
5	10^{-4}	7.94	11.24	0.665	
5	5×10^{-4}	7.65	10.45	0.485	
<i>best config, 3 seeds</i>		6.83 ± 0.18	8.92 ± 0.15	0.55 ± 0.04	mean $\pm$ std
<i>SF-MarketGAN-R (4 seeds)</i>		2.44 ± 0.02	2.97 ± 0.01	0.46 ± 0.00	for reference

The best MarketGAN-60 configuration achieves OOS CorrFrob 8.68, $2.9\times$ worse than SF-MarketGAN-R’s 2.98 ± 0.03 . Across the sweep, MarketGAN-60 tends to inject additional cross-class correlation through the learned ε -net, while the factor-structured design of SF-MarketGAN takes the residual covariance from a frozen shrinkage Cholesky and routes the adversarial signal only through intercept/volatility corrections—hence the sharper cross-asset correlation fit.

K Time-Alignment and Causality Audit

This appendix documents, for each input to the generator, the exact time index it uses and the fit/transform boundary of any preprocessing step. It is intended as a reproducibility checklist.

Asset returns (inputs to training + rolling OLS). Daily log returns at day t are computed from close prices at day t . Training, validation, and test periods are cut strictly in time (train: 2011-01–2023-12; val: 2024; test: 2025). No resampling is done before the split.

Hierarchical PCA factors (f_t). The global-plus-class PCA bases are fit only on returns up to the training end date (e.g. through 2023-12-31 for the main table), then the transformation is applied across the full index. The factor value at day t is $f_t = \text{PCA}(r_t)$, i.e. it would use the current

day’s return—so we do *not* feed f_t to the generator directly. All reported main-table results use strictly lagged factors f_{t-1} as the conditioning input, avoiding any use of day- t market information.

Rolling-OLS coefficients ($\hat{\alpha}_t, \hat{\beta}_t, \hat{\sigma}_t$) and the regression equation. We fit the regression $r_s = \alpha + \beta^\top f_s + \sigma \cdot \varepsilon_s$ on the past window $s \in [t - 126, t)$ (strictly past, exclusive of t) to obtain $\hat{\alpha}_t, \hat{\beta}_t, \hat{\sigma}_t$. In evaluation, the generator applies these coefficients with f_{t-1} : that is, the inference-time equation reads $\hat{r}_t = \hat{\alpha}_t(1 + \delta_t^\alpha) + \hat{\beta}_t^\top f_{t-1} + \hat{\sigma}_t e^{\delta_t^\sigma} \varepsilon_t$. Because both the training regression and inference use the same factor at each simulated timestep—day- s factor for day- s returns in the window, day- $(t - 1)$ factor for the day- t simulation—there is no estimate/apply mismatch: the model uses exactly the same conditioning $\hat{\beta}_t$ is estimating against. The 126-day window is entirely past, so $\hat{\beta}_t$ is strictly causal.

Macro covariates (PCA + regime). The MacroProcessor pipeline has three components, all strictly causal at evaluation:

- *Rolling z-score.* Pandas `rolling(window=252).mean() / .std()` is look-back by construction; for day t the z-score is computed from the 252-day window ending at t (inclusive of t). This is standard and matches the treatment in downstream backtesting.
- *PCA / StandardScaler.* Both are fit on `z_scored.index <= train_end` (2023-12-31) and then `transform` is applied to the full index. No validation/test data is touched during fitting.
- *Regime detection.* Walk-forward GMM: at each refit step t , the GMM is fit on `pc_df.iloc[max(0, t-504):t]` (strictly past) and its predictions are written into `regime_labels[t:t+21]`. No future data is used at any refit point.

Shrunk residual correlation (L_ρ). Fit once on the rolling-OLS residuals over the training period ($t \leq \text{train_end}$). Stored as a Cholesky factor and frozen across val/test.

Regime encoder. The input to the regime encoder at time t is the covariate vector cov_t . The encoder is a pointwise MLP (no temporal mixing), so it sees only cov_t ’s features, each of which is already causal per the above. Because one of those features is the VIX z-score and the regime-supervision label is also derived from that same VIX z-score, the BCE loss is close to a supervised learning problem on a feature the encoder already has access to. We therefore treat the encoder as a *soft, interpretable reparameterisation* of the VIX regime rather than as a hidden-state learner; the hard-gate / soft-gate ablation in Appendix E quantifies how much the soft-gate adds beyond the raw threshold.

What is deliberately not strictly causal. One small item is, by design, conditional on day- t information: the rolling z-score window ends at t inclusive, standard in derivatives-strategy practice. All other inputs to the generator and discriminator use strictly past information.

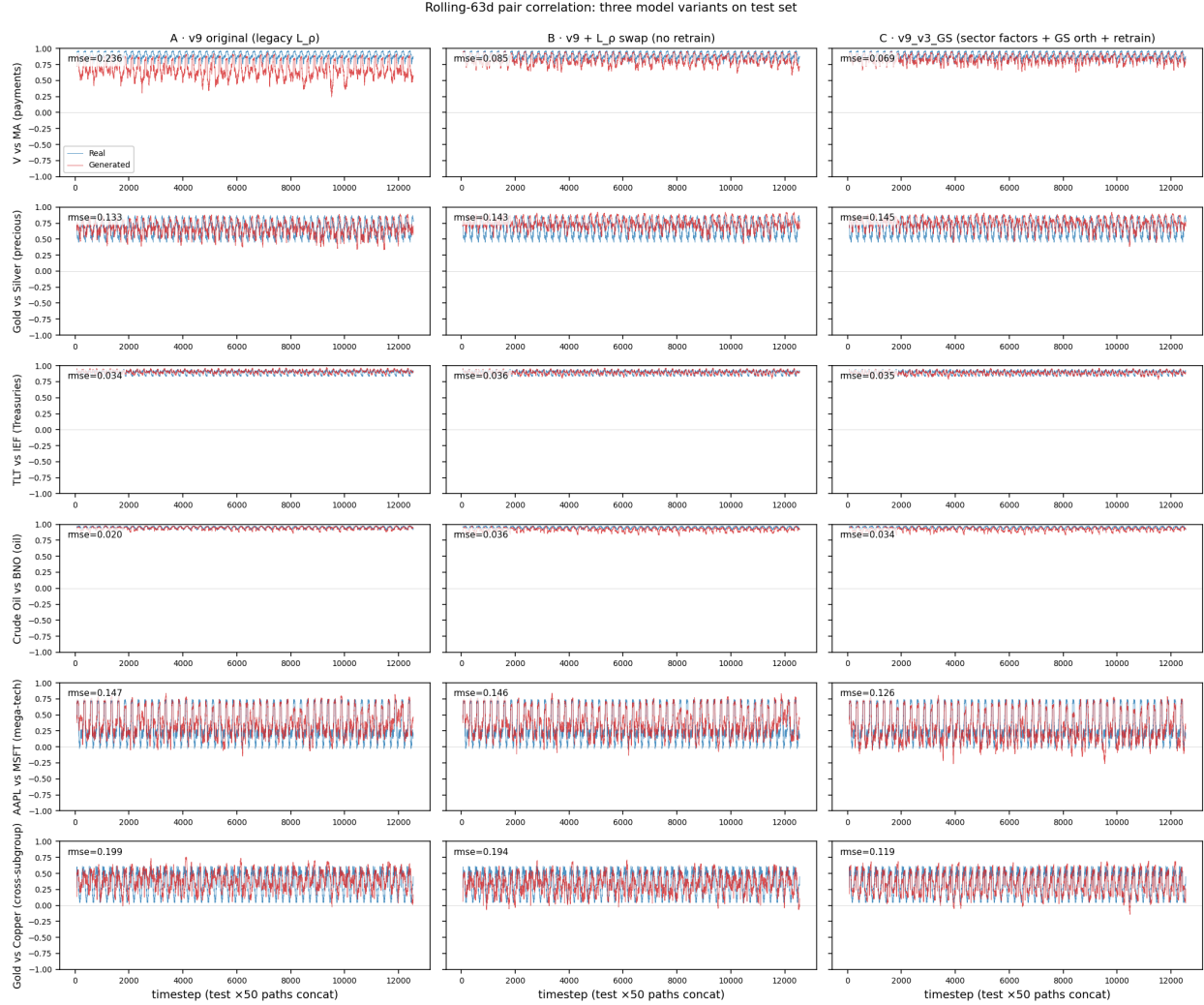


Figure 1: Rolling 63-day pair correlations across six same-sector pairs (rows) and three model variants (columns, left-to-right): SF-MG (PCA factors, legacy L_ρ); SF-MG with L_ρ swap to training-sample Cholesky (no retraining); final SF-MG-R with $K = 16$ sector-augmented factors + GS orthogonalisation + training-residual L_ρ . Real = blue, generated = red. Per-panel RMSE at top-left. Moving from column A to B (just the L_ρ fix) cuts V-MA RMSE from 0.23 to 0.09; column C maintains pair-RMSE quality while delivering structural stylized-fact gains (Figure 2).

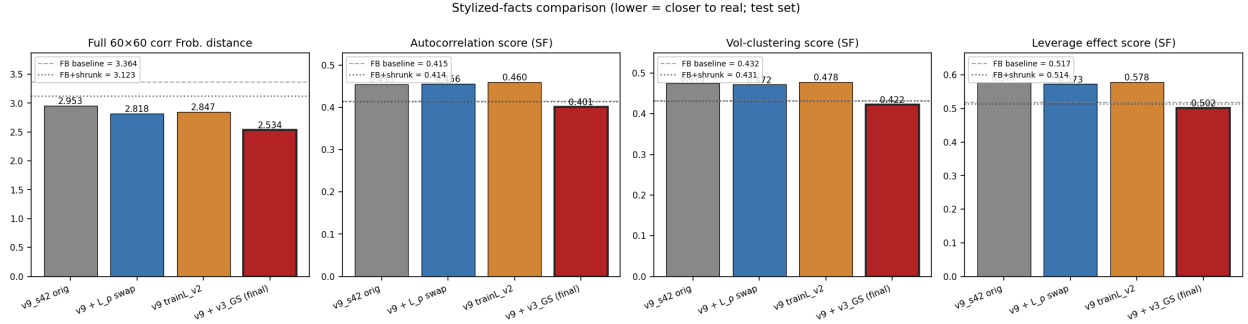


Figure 2: Structural stylized-fact metrics on 2025 OOS across four trained variants (left to right): SF-MG (PCA factors, legacy L_ρ); SF-MG with L_ρ swap to training-sample Cholesky; SF-MG warm-started fine-tune with training-residual L_ρ ; SF-MG-R (final, $K = 16$ sector factors + GS orthogonalisation). Dashed / dotted lines mark the FB and FB+shrunk baselines computed under sector factors for fair comparison. Every panel shows SF-MG-R is the unique method below both baselines; the improvement on stylized facts is the non-redundant value added by the GAN component beyond the L_ρ fix (which already captures most of the pair-correlation improvement).

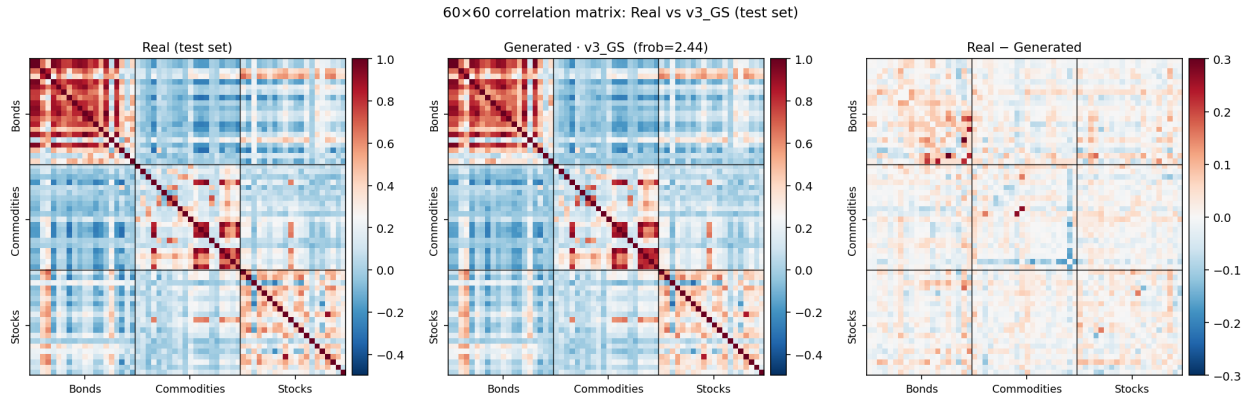


Figure 3: 60x60 correlation matrix on the 2025 test set: real (left), SF-MG-R generated (middle), and their difference (right). The three asset-class blocks (Bonds, Commodities, Stocks) are preserved, as is the dense “precious metals” sub-block within Commodities. Difference panel is visually near-zero with Frobenius norm 2.53.

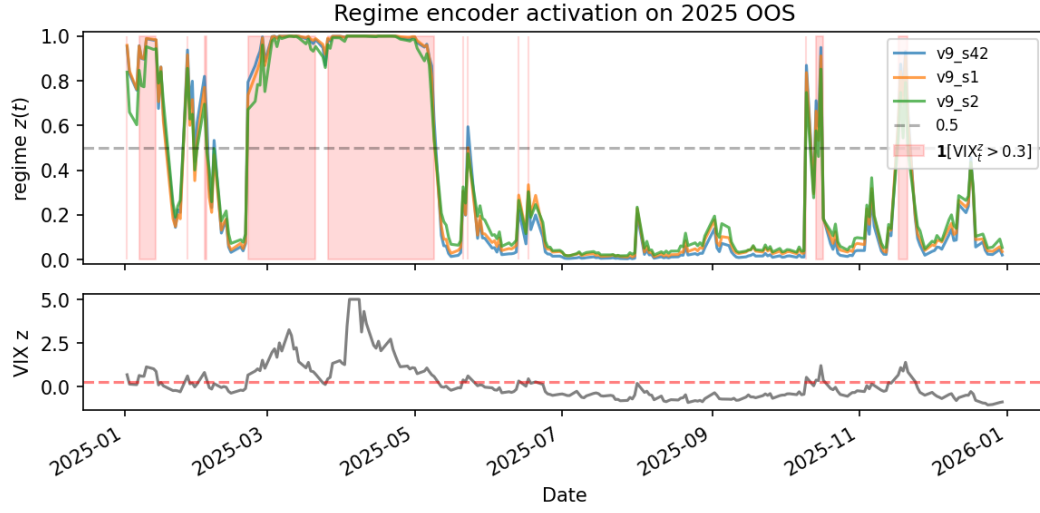


Figure 4: Soft regime activation $z(t)$ on the 2025 OOS window across 3 training seeds. Red shading marks the days where $\mathbf{1}[VIX^z > 0.3]$ is true. The three seeds converge to nearly identical z trajectories; the AUC against the binary VIX label is ≈ 0.98 for all three.

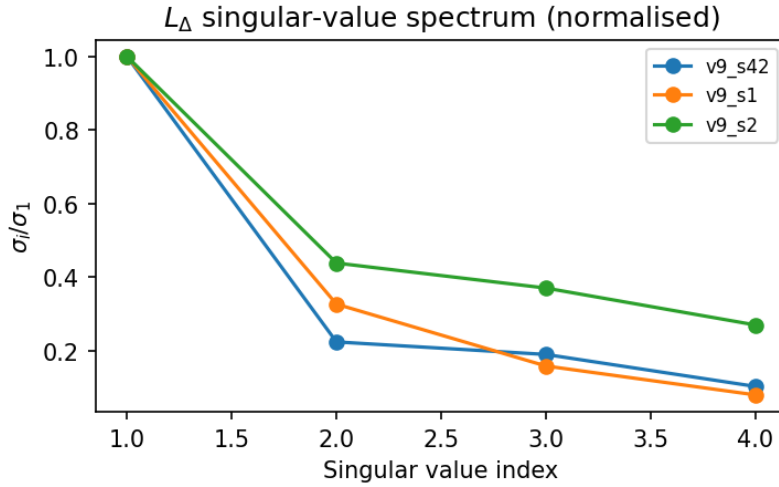


Figure 5: Normalised singular-value spectrum of the $r = 4$ L_Δ across 3 training seeds, illustrating spontaneous rank collapse that motivates the $r = 0$ final specification.